
PROVISIONAL PATENT APPLICATION

System and Method for a Tiered Artificial Intelligence Compute Token Framework for Query Complexity Control, Consumption Metering, Role-Based Pricing, and Environmentally Efficient AI Delivery

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Conception Date: June 12, 2026 — The inventor conceived the tiered AI token framework during active development of the Treasury Risk Monitor AI platform, the earliest components of which bear file timestamps of June 12, 2026. The token system concept arose directly from the inventor's practical experience architecting agentic AI queries and observing the compute efficiency difference between structured agentic queries and open-ended LLM prompts. This provisional application is filed on July 15, 2026.

ABSTRACT

A system and method for a tiered artificial intelligence (AI) compute token framework that controls, prices, meters, and routes AI query workloads across three defined complexity tiers. Three token tiers — Basic, Mid, and Advanced — correspond to increasing query complexity, compute requirements, and price points. A Token Metering Engine classifies incoming queries, validates the user's purchased tier entitlement, gates or permits execution, and routes the query to the appropriate compute path. A Token Ledger maintains an auditable consumption record. The framework supports consumer application credits, enterprise seat licensing, and LLM platform licensing. By routing simple queries to lightweight model paths, the system structurally reduces unnecessary high-compute AI invocations, resulting in measurable reductions in data center load, power consumption, and carbon footprint. Conceived June 12, 2026.

FIELD OF THE INVENTION

The present invention relates to artificial intelligence systems, and more particularly to a system and method for classifying, metering, pricing, and routing AI query workloads using a tiered token framework applicable to large language models, multi-agent AI systems, and AI-powered application platforms.

BACKGROUND

The rapid proliferation of large language models (LLMs) and AI-powered applications has created significant challenges in compute resource management, cost allocation, and environmental impact. Current AI billing models treat all queries uniformly, invoking full model infrastructure regardless of task complexity. A simple factual lookup consumes the same infrastructure pathway as a complex multi-step research and code generation task, resulting in unnecessary energy consumption, disproportionate data center load, limited monetization options, and lack of structured access control tied to user role or query complexity. No widely adopted system exists that classifies AI queries by complexity, routes them to appropriately scaled compute paths, enforces tier-based access through economic token mechanisms, and produces auditable consumption records.

CONTEXT OF CONCEPTION

The inventor, John Kirshy, conceived this invention on or about June 12, 2026 while developing the Treasury Risk Monitor — a multi-agent AI fraud detection platform built on IBM watsonx. During that development, the inventor directly observed that structured agentic AI queries (queries constrained to a defined task scope and pattern) required significantly less LLM compute than open-ended generative prompts of equivalent apparent complexity. This observation led to the conception of a formal tiered token system that could standardize, price, and enforce this compute efficiency difference across any AI platform or application.

SUMMARY OF THE INVENTION

The present invention provides a Tiered AI Compute Token Framework comprising three token tiers, a Token Metering Engine, a Tier Router, a Token Ledger, and a licensing model applicable across consumer, enterprise, and LLM platform markets. The framework enables differentiated pricing by query complexity, role-based access control, consumption metering, and structural reduction of unnecessary AI compute consumption.

In one aspect, the invention provides a system comprising:

- **A three-tier token classification schema** defining Basic, Mid, and Advanced token types mapped to query complexity levels.
- **A Token Metering Engine** that classifies queries, validates tier entitlement, gates execution, and routes to the appropriate compute path.
- **A Token Ledger** that maintains an auditable, real-time record of token consumption per user and enterprise.
- **A Licensing Framework** applicable to consumer app credits, enterprise seat licensing, and LLM platform licensing.
- **An environmental efficiency mechanism** by which simple queries are structurally prevented from invoking high-compute model infrastructure.

DETAILED DESCRIPTION OF THE INVENTION

1. The Three Token Tiers

The invention defines three token types:

Tier	Token Type	Query Scope	Compute Path	Primary User
1	Basic Token	Single-turn, simple factual queries	Lightweight / rule-based path	General consumers
2	Mid Token	Multi-turn sessions, explanations, moderate analysis	Standard LLM inference	Power users, analysts, SMB
3	Advanced Token	Complex research, infrastructure provisioning, agentic pipelines	Full LLM + agentic orchestration	Developers, enterprise architects

Each tier carries a distinct price point, compute pathway, and scope of permitted query complexity. Users purchasing Tier 1 (Basic) tokens receive access only to simple single-turn query execution on a lightweight model path. Users purchasing Tier 3 (Advanced) tokens receive full LLM inference, multi-agent orchestration, and infrastructure generation capabilities. Users who attempt queries beyond their purchased tier are gated by the Token Metering Engine and prompted to acquire a higher tier. The economic design creates natural incentives for users to self-select the appropriate tier, reducing wasteful over-provisioning.

2. Token Metering Engine

The Token Metering Engine is positioned between the user or application interface and the LLM or agent execution environment. It performs: (1) Classification — analyzes the incoming query against the three tier definitions; (2) Entitlement Validation — compares the classified tier against the user's purchased token tier; (3) Gating — blocks execution if complexity exceeds purchased tier; (4) Routing — directs the approved query to the designated compute path; (5) Ledger Recording — logs tier, timestamp, user identifier, and consumption.

Note: The specific algorithmic implementation of query complexity classification within the Token Metering Engine is intentionally described at the functional level in this provisional application to preserve the broadest possible claim scope.

3. Token Ledger

The Token Ledger maintains a persistent, auditable record of all token consumption events. Each record comprises: user or enterprise identifier, token tier consumed, query timestamp, compute path invoked, and remaining token balance. Supports enterprise compliance reporting, environmental impact auditing, and billing reconciliation.

4. System Architecture

Component	Role
User / Application	Initiates query; holds purchased token tier credential
Token Classification Interface	Presents tier options or reads purchased tier from credential
Token Metering Engine	Classifies query complexity; validates tier entitlement; gates or allows; logs to ledger
Tier Router	Directs approved query to appropriate compute path
LLM / Agent Execution Layer	Executes query at compute level matched to token tier
Token Ledger	Maintains auditable record of consumption per user, tier, and timestamp
Deliverable / Response	Returns output; updates token balance

5. Licensing Framework

- **Consumer Application Credits:** AI-powered application developers integrate the token framework to sell tiered credit packages. Tier 1 credits serve as entry products; Tier 3 credits are premium. Users purchase additional credits as consumption requires.
- **Enterprise Seat Licensing:** Organizations license the framework per seat or department. Developer roles receive Tier 3; general staff receive Tier 1 or 2. Pricing scales by tier, seat count, and volume.
- **LLM Platform Licensing:** The tier framework is offered to LLM providers as a standardized monetization and consumption-control layer, enabling differentiated product offerings and a credible, structurally enforced green AI market position.

6. Environmental Efficiency and Agentic AI Connection

By classifying queries at intake and routing Tier 1 queries exclusively to lightweight model paths, the system structurally prevents simple queries from invoking full LLM infrastructure. The Token Ledger produces auditable data quantifying compute reduction by tier. Furthermore, structured agentic AI queries — where queries are constrained to defined task patterns rather than open-ended generative prompts — reduce the compute burden on the underlying LLM relative to unbounded prompting. This

agentic efficiency is an additional environmental benefit of the tiered framework and was directly observed by the inventor during development of the Treasury Risk Monitor beginning June 12, 2026.

CLAIMS

What is claimed is:

1. A computer-implemented system for tiered artificial intelligence compute token management, comprising: a three-tier token classification schema defining a first tier for simple single-turn queries, a second tier for multi-turn sessions and moderate analysis, and a third tier for complex research, infrastructure generation, and agentic pipeline execution; a Token Metering Engine configured to receive an incoming AI query, classify the query against the three-tier schema, validate the user's purchased token tier entitlement, gate or permit execution based on entitlement validation, and route the query to a compute path corresponding to the validated tier; a Token Ledger configured to record consumption events per user, tier, and timestamp; and a pricing model wherein each token tier carries a distinct price point.
2. The system of claim 1, wherein the Token Metering Engine routes Tier 1 queries to a lightweight or rule-based model path, Tier 2 queries to a standard LLM inference path, and Tier 3 queries to a full LLM with multi-agent orchestration.
3. The system of claim 1, wherein the Token Ledger produces auditable consumption records suitable for enterprise compliance reporting and environmental impact quantification.
4. The system of claim 1, further comprising a consumer application licensing interface through which third-party AI application developers offer tiered token credits to end users.
5. The system of claim 1, further comprising an enterprise seat licensing interface through which organizations assign token tier entitlements to users based on role, wherein developer roles receive Tier 3 entitlements and general users receive Tier 1 or Tier 2 entitlements.
6. The system of claim 1, further comprising an LLM platform licensing interface through which the tiered token framework is offered to large language model providers as a monetization and compute-efficiency layer.
7. A method for routing artificial intelligence queries by complexity tier, comprising: receiving a query from a user; classifying the query into one of a first, second, or third complexity tier; validating the user's purchased token tier against the classified tier; routing the query to a compute path corresponding to the validated tier; and recording the consumption event in a token ledger.
8. The method of claim 7, further comprising blocking execution and notifying the user when the classified query tier exceeds the user's purchased token entitlement.
9. The system of claim 1, wherein structuring AI queries as purpose-built agentic tasks constrained to a defined scope reduces compute consumption on the underlying LLM relative to open-ended

generative prompts of equivalent apparent complexity, and wherein the tiered framework captures and enforces this efficiency difference through tier routing.

10. A non-transitory computer-readable medium storing instructions that, when executed, implement the system of claim 1, wherein the invention was conceived on June 12, 2026.

BRIEF DESCRIPTION OF FIGURES

Figure 1 — Overall System Architecture: User -> Token Classification Interface -> Token Metering Engine -> Tier Router -> LLM/Agent Execution -> Token Ledger -> Deliverable.

Figure 2 — Three Tier Token Schema: Tier 1/2/3 mapped to query scope, compute path, pricing, and user role.

Figure 3 — Token Metering Engine Flow: Classify -> Validate Entitlement -> Gate/Allow -> Route -> Log -> Respond.

Figure 4 — Token Ledger Schema: User ID, Tier, Timestamp, Compute Path, Tokens Consumed, Remaining Balance.

Figure 5 — Licensing Markets: Consumer App Credits / Enterprise Seat Licensing / LLM Platform Licensing.

Figure 6 — Environmental Efficiency: Compute reduction by tier — Tier 1 lightweight path only; Tier 3 full agentic pipeline.

Figure 7 — Conception Context: Treasury Risk Monitor development timeline June-July 2026 as context for invention conception.

INVENTOR DECLARATION

I, John Kirshy, hereby declare that I am the original inventor of the subject matter described in this provisional patent application. I conceived this invention on or about June 12, 2026 during active development of the Treasury Risk Monitor AI platform. I believe the invention is new and was conceived by me independently. I acknowledge the duty to disclose to the USPTO all information known to me to be material to patentability.

Signature: _____ Date: July 15, 2026

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